All Five 'Letters' of DNA Found in an Ancient Asteroid

From: ScienceDaily Strange & Offbeat Mined: 2026-11-05 5 min read

All five of the molecular "letters" of DNA and RNA have been found in a 4.5-billion-year-old asteroid. A new study reports that all five fundamental **nucleobases** have been detected in samples from the asteroid Ryugu, which Japan's Hayabusa2 mission brought back to Earth in 2020. The study, published in the journal Nature Astronomy, was carried out by a team of Japanese scientists. **Nucleobases** are nitrogen-containing organic molecules that spell out genetic information in DNA and RNA. The five main ones are adenine and guanine, known as **purines**, and cytosine, thymine and uracil, known as **pyrimidines**. They combine with sugars and phosphates to make **nucleotides**, the building blocks of genetic material. Without them, the genetic code that lets organisms grow, reproduce and evolve would not exist. This is not the first time scientists have spotted these molecules in space rock. In 2023, an international team reported finding uracil, one of the five, in the same Ryugu samples. Meteorites that fell to Earth have also yielded **nucleobases**, and samples from NASA's mission to the asteroid Bennu yielded all five in 2025. But meteorites that land on Earth can be **contaminated** along the way. The Ryugu samples are different: Hayabusa2 delivered 5.4 grams of **pristine** material, collected in space and protected from Earth's contamination, so researchers could study it in ultra-clean labs. They extracted the organic molecules using water and hydrochloric acid, then purified them for testing, and found all five **nucleobases** in both samples they analysed, in roughly similar amounts. The results suggest that the ingredients for life may have been widespread throughout the early Solar System. Asteroids like Ryugu are leftovers from the Solar System's early days. Together, the findings suggest that key parts of genetic material may have formed in space and later been delivered to the early Earth.

Word Treasure

nucleobases -- Nucleobases are the tiny molecules that spell out genetic instructions.
purines -- Purines are two of the five nucleobases: adenine and guanine.
pyrimidines -- Pyrimidines are three of the five nucleobases: cytosine, thymine and uracil.
pristine -- Pristine means clean and untouched.
nucleotides -- Nucleotides are the building blocks of DNA and RNA.
contaminated -- Contaminated means dirtied by something unwanted.

Background you need

DNA and RNA are the instruction books inside living things that tell them how to grow.

An asteroid is a rocky object that orbits the Sun, left over from the early Solar System.

A space mission can bring samples back to Earth so scientists can study them in clean labs.

Finding the same building blocks of life in space rocks suggests those ingredients were common in the early Solar System.

Break it down

LEAD: All five DNA letters found in an ancient asteroid
KEY EVENT: Japanese scientists confirm all five nucleobases in Ryugu samples
+- WHO: Japanese scientists and JAXA's Hayabusa2
+- WHERE: Samples from the asteroid Ryugu
+- WHEN: Returned to Earth in 2020, study published now
EVIDENCE: 5.4 grams of pristine material studied in ultra-clean labs
OPEN QUESTION: Did asteroids deliver life's ingredients to early Earth?

Why it matters

If the building blocks of life came from space, the story of life on Earth is part of a much bigger cosmic story.

Test what you remember

1. How many nucleobases were found in the Ryugu samples?

- (A) Four
- (B) Only one
- (C) All five
- (D) Three

2. Which spacecraft brought the Ryugu samples to Earth?

- (A) China's Chang'e mission
- (B) Japan's Hayabusa2
- (C) The International Space Station
- (D) NASA's Bennu mission

3. In what year did Hayabusa2 bring the samples back?

- (A) 2020
- (B) 2025
- (C) 2019
- (D) 2023

4. What are nucleobases?

- (A) The molecular letters of DNA and RNA
- (B) Tiny pieces of rock
- (C) Types of stars
- (D) Parts of a spacecraft

5. How much pristine asteroid material did Hayabusa2 deliver?

- (A) 540 grams
- (B) 5.4 kilograms
- (C) 54 grams
- (D) 5.4 grams

6. Why are the Ryugu samples special compared with meteorites?

- (A) They fell in Australia
- (B) They are much larger
- (C) They contain gold
- (D) They were collected in space and protected from Earth's contamination

Answer key (try the quiz first!): 1: C 2: B 3: A 4: A 5: D 6: D

Questions to think about

MY THOUGHTS (at least 20 words):